Quantification: 699_HYP_HP_z25c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	15
interior lacuna count	11
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1867.27	1302	2262.32	px^2
major_axis_length_px	lacuna shape	83.6663	67.6703	57.6428	px
minor_axis_length_px	lacuna shape	31.6216	30.4825	16.4785	px
aspect_ratio	lacuna shape	2.8735	2.348	1.8794	unitless
eccentricity	lacuna shape	0.8795	0.9048	0.0863	unitless
solidity	lacuna shape	0.7821	0.7724	0.1042	unitless
roots_count	headline	5.0909	6	3.0151	count
ring_length_r30_px	headline	218.546	229	101.164	px
ring_length_r60_px	reported	441.818	580	211.467	px
owned_length_px	ownership-dependent	575.335	318.38	721.268	px
edge_count	ownership-dependent	29.9091	14	40.7859	count
mean_edge_length_px	ownership-dependent	21.2256	22.3311	9.5514	px
perimeter_px	normalised	262.945	218.392	198.628	px
ring_area_r30_px2	normalised	8366.91	6473	5580.12	px^2
ring_area_r60_px2	normalised	20013.5	17377	11010.8	px^2
in_frame_fraction_r30	normalised	0.9852	1	0.0489	unitless
in_frame_fraction_r60	normalised	0.9732	1	0.089	unitless
ring_density_r30	normalised	0.0294795	0.0282713	0.0145203	px^-1
ring_density_r60	normalised	0.0245628	0.0267594	0.0131173	px^-1
roots_per_100px_perimeter	normalised	2.2866	1.7645	1.6913	per 100 px
ring_attached_length_r30_px	network v2	169.364	175	87.5332	px
ring_attached_length_r60_px	network v2	302.454	271	161.716	px
ring_length_w_r30_px	network v2	245.972	253.652	113.457	px
ring_length_w_r60_px	network v2	494.851	637.931	234.938	px
sholl_crossings_r10	network v2	5.1818	4	3.1565	count
sholl_crossings_r20	network v2	7.0909	7	4.5487	count
sholl_crossings_r30	network v2	5.6364	6	2.3355	count
loop_count	loops, report only	1.9091	0	4.9082	count
loop_edge_share	loops, report only	0.0655	0	0.153	share
ring_width_mean_r30_px	width	7.1096	6.8704	1.1017	px

Per field
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measure	value	units
analysed_area_px2	1.00613e+06	px^2
total_skeleton_length_px	16953	px
canalicular_length_density_per_px	0.0168497	px^-1
junction_count	337	count
junction_density_per_px2	0.00033495	px^-2
lacuna_count	15	count

lacunae_per_px2	1.491e-05	px^-2
skeleton_component_count	320	count
mean_component_length_px	52.9781	px
median_component_length_px	17	px
field_length_density_w_per_px	0.018819	px^-1
field_density_without_flagged_per_px	0.0150271	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	767	px
unattached_component_count	38	count
canalicular_length_density_attached_per_px	0.0168497	px^-1
field_length_density_with_unattached_per_px	0.017612	px^-1
loop_count	35	count
loop_edge_share	0.1371	share
edge_count	985	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.0189	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10.099	px
width_pixels_used	12176	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	462	31.514	19.2261	1.6391	0.7923	0.913	1.2654	8.9567	716.093	True
2	616	37.3985	23.34	1.6023	0.7814	0.8415	0.9265	217.943	405.003	False
3	1197	51.8325	30.7893	1.6835	0.8045	0.9089	-1.2998	286.212	636.439	False
4	1418	83.5326	24.5778	3.3987	0.9557	0.7698	-1.1714	319.255	630.773	False
5	4490	102.405	66.4904	1.5401	0.7605	0.7426	0.7875	384.318	34.5902	True
6	10182	211.16	88.8809	2.3758	0.9071	0.7109	0.0099	535.028	34.0221	True
7	1476	60.9018	39.0949	1.5578	0.7668	0.7644	-0.4297	540.3	441.844	False
8	1461	75.7643	32.2675	2.348	0.9048	0.7334	-1.4732	556.736	495.522	False
9	1302	67.6703	25.7149	2.6316	0.925	0.8906	1.4126	532.052	690.266	False
10	950	46.9475	30.4825	1.5401	0.7605	0.7724	-1.05	698.178	753.763	False
11	6771	137.82	71.3934	1.9304	0.8554	0.8796	-1.1154	815.976	973.564	True
12	675	96.7006	11.9848	8.0686	0.9923	0.7196	-0.8157	830.723	702.28	False
13	8532	242.448	76.4179	3.1727	0.949	0.5319	-1.433	856.199	279.575	False
14	2265	113.598	32.1074	3.5381	0.9592	0.8015	1.0532	904.727	521.268	False
15	648	43.5355	21.061	2.0671	0.8752	0.8686	-1.0803	997.846	882.89	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	1	74	247	112.255	5	22.451	4.7104
2	6	229	669	2519.64	138	18.2582	7.8237
3	0	74	242	0	0	0	6.069
4	6	183	427	210.095	12	17.508	6.055

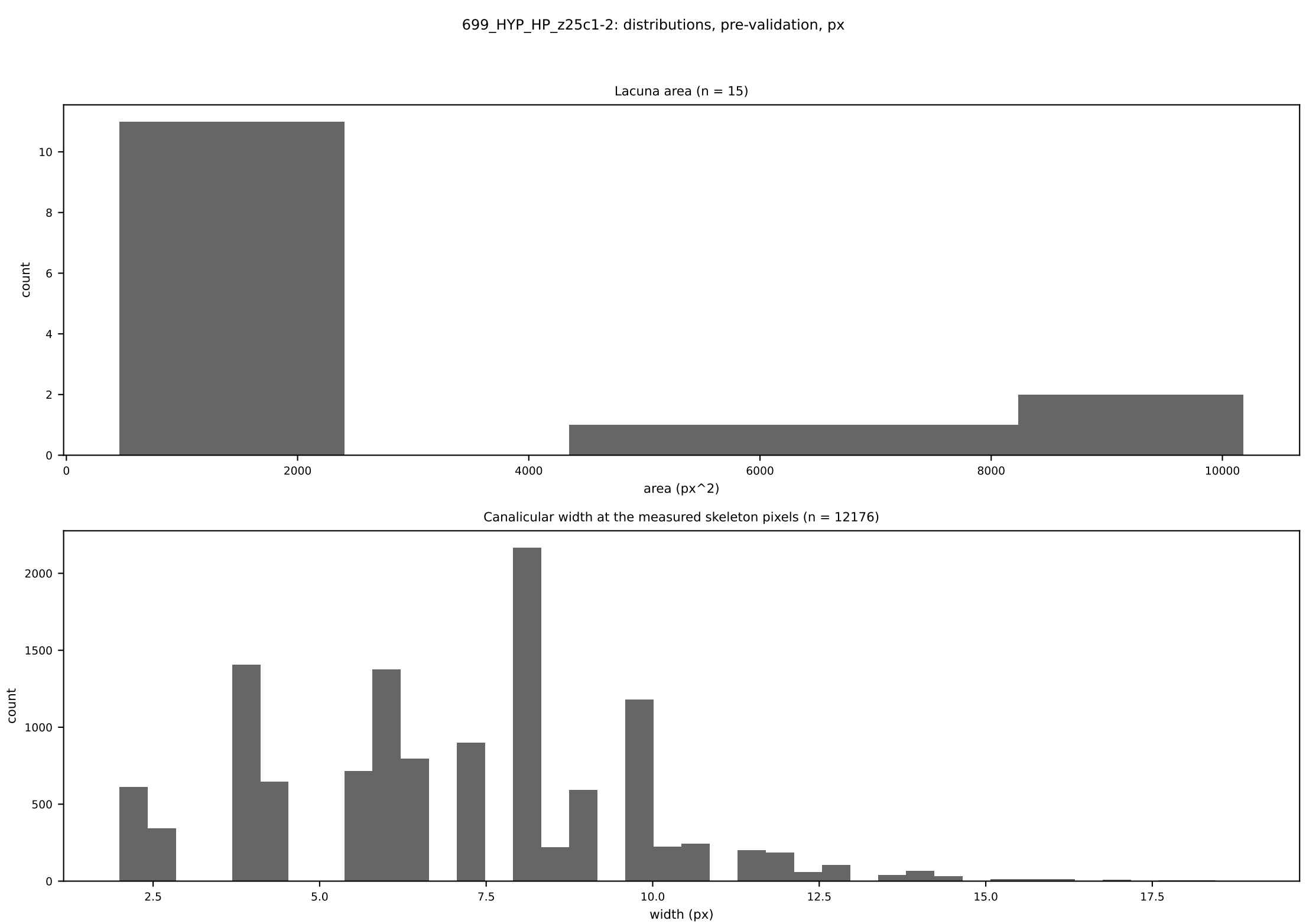
5	10	322	610	1862.73	107	17.4087	9.1732
6	5	179	245	125.255	7	17.8935	9.8643
7	5	318	580	545.546	24	22.7311	8.304
8	6	309	629	1193.95	70	17.0564	7.4716
9	2	299	583	318.38	14	22.7414	6.8704
10	9	274	584	552.789	33	16.7512	6.1521
11	0	14	47	0	0	0	5.7599
12	8	179	260	248.492	8	31.0616	7.1215
13	9	355	623	424.291	19	22.3311	6.5385
14	3	103	168	278.291	10	27.8291	9.5317
15	2	81	95	37.2132	1	37.2132	6.2677

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	91.2548	3226	9227	0.6074	0.5675	0.0229386	0.0267693	1.0958
2	125.539	5894	17377	1	1	0.0388531	0.0384992	4.7794
3	150.167	4208	10231	1	1	0.0175855	0.0236536	0
4	231.421	6473	15957	1	1	0.0282713	0.0267594	2.5927
5	467.688	7807	15429	0.7054	0.654	0.041245	0.0395359	2.1382
6	694.642	11969	22060	0.6372	0.5918	0.0149553	0.0111061	0.7198
7	282.806	6457	15698	1	1	0.0492489	0.0369474	1.768
8	340.049	6155	14117	1	1	0.0502031	0.0445562	1.7645
9	169.924	7529	20703	1	1	0.0397131	0.0281602	1.177
10	163.924	6994	18029	1	1	0.0391764	0.0323923	5.4904
11	520.286	10251	22904	0.728	0.6854	0.00136572	0.00205204	0
12	218.392	8675	21397	1	1	0.020634	0.0121512	3.6631
13	823.009	24482	51149	1	1	0.0145004	0.0121801	1.0935
14	270.593	10058	22934	1	1	0.0102406	0.00732537	1.1087
15	116.569	5111	12557	0.8377	0.7049	0.0158482	0.0075655	1.7157

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	50	82	83.5269	277.551	2	2	3
2	164	498	253.652	761.097	4	8	9
3	24	77	82.1838	270.137	1	3	4
4	143	271	204.681	471.191	6	5	6
5	303	568	373.262	703.269	7	7	7
6	97	117	200.196	272.48	3	6	5
7	317	453	359.421	650.316	9	11	7
8	274	462	343.208	708.63	9	10	6
9	194	341	340.149	637.931	4	17	7
10	237	497	305.551	652.144	8	9	7
11	0	0	14.2426	51.2132	0	1	1
12	175	253	211.693	309.747	3	3	4
13	176	269	398.09	685.671	9	7	8
14	97	124	119.296	192.894	3	3	2
15	62	82	87.7696	103.598	1	2	2


699_HYP_HP_z25c1-2: distributions, pre-validation, px
Lacuna area (n = 15)
count
area (px^2)
0
2
4
6
8
10
0
2000
4000
6000
8000
10000
Canalicular width at the measured skeleton pixels (n = 12176)
count
width (px)
0
500
1000
1500
2000
2.5
5.0
7.5
10.0
12.5
15.0
17.5